Stars

How many stars are there?

Classification of Stars

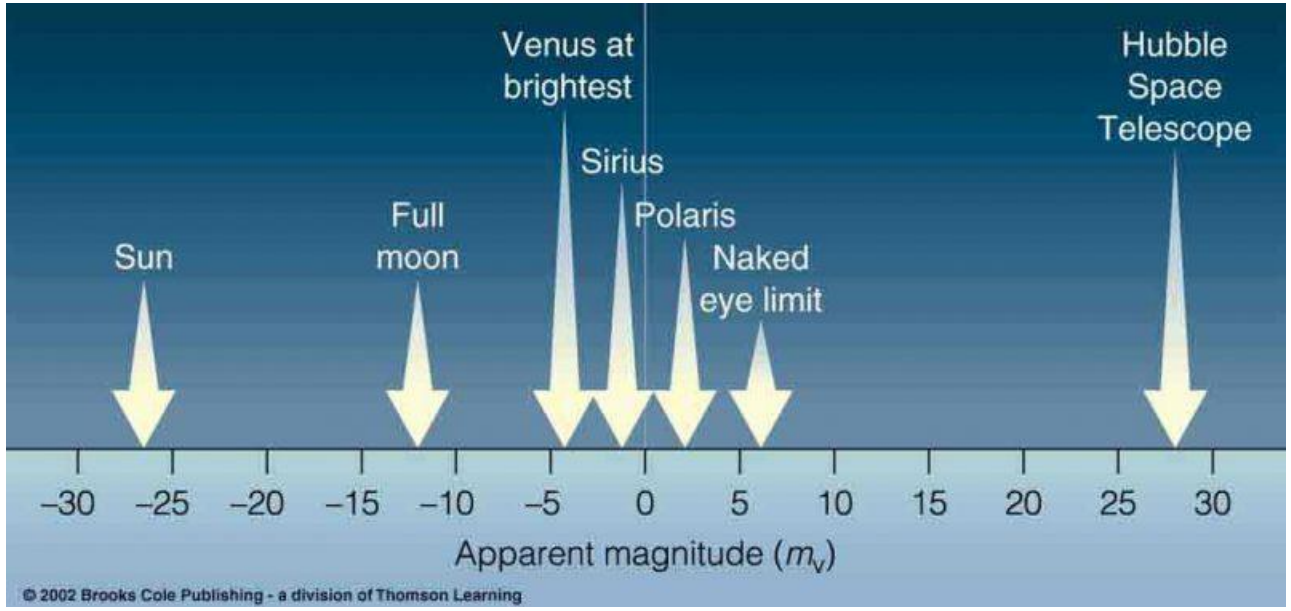
- **Magnitude**
- **Luminosity**
- **Temperature**
- Chemical Composition
- Mass

Apparent Magnitude

- Apparent magnitude is how bright a star **appears** in the sky
- Measured on a scale where brightness **decreases** as the numbers **increase**
- Therefore the brightest stars have a **negative** magnitude

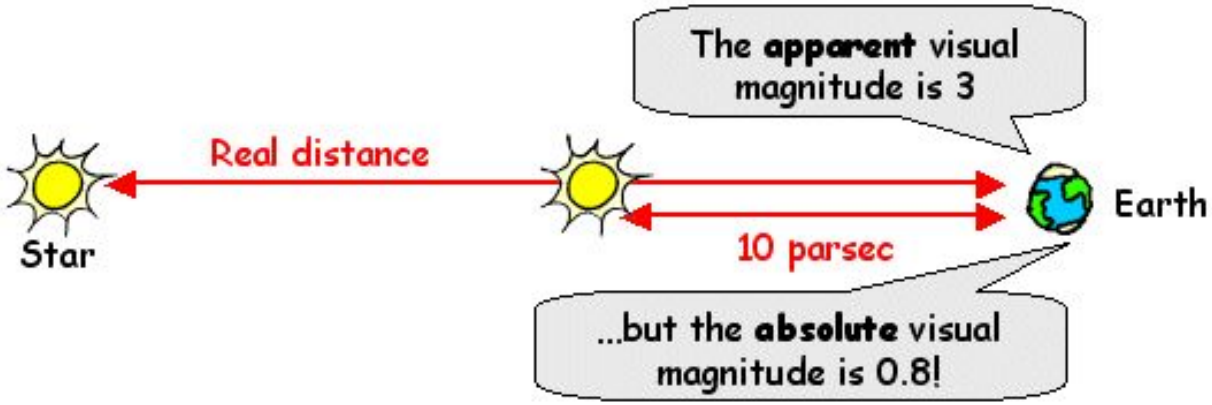
Apparent Magnitude

- However, it does not indicate the actual **brightness** of a star because it does not account for **distance**

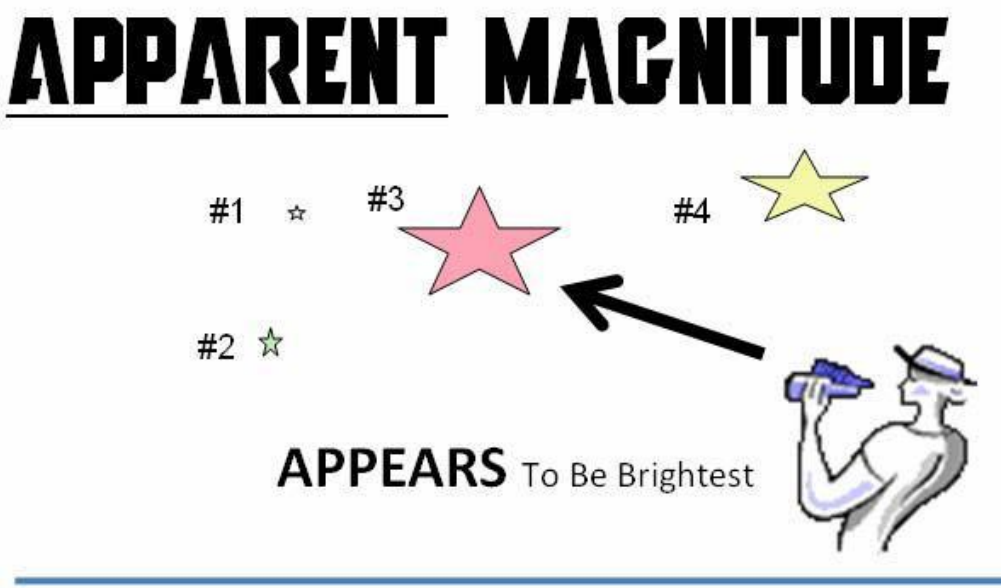


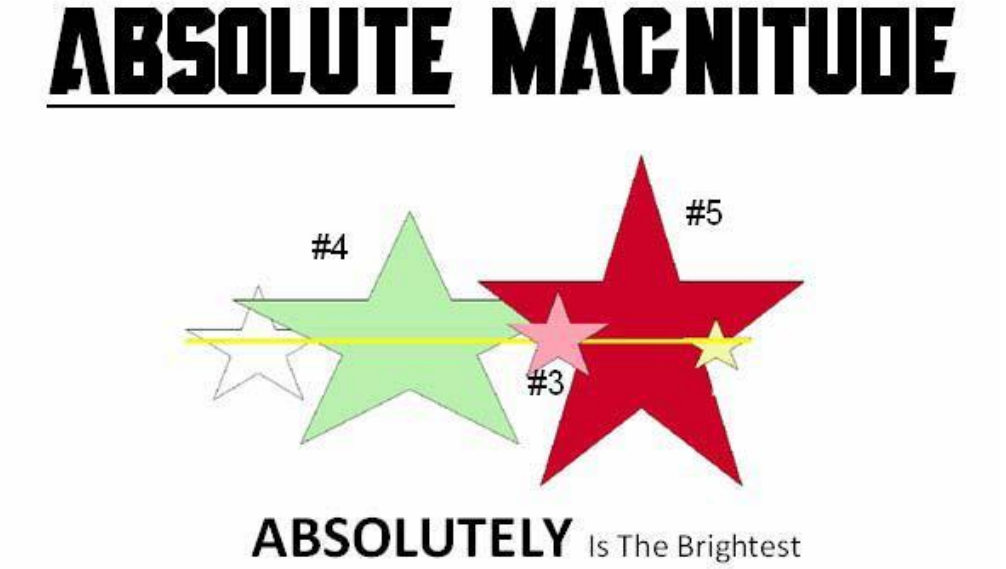
Absolute Magnitude

- Accounts for **distance**
- Measures how bright an object would appear if it were **33 light years** from Earth
- Allows you to compare stars as if they were the **same** distance away



Magnitude


APPARENT MAGNITUDE
#1
#3
#4
#2
APPEARS To Be Brightest

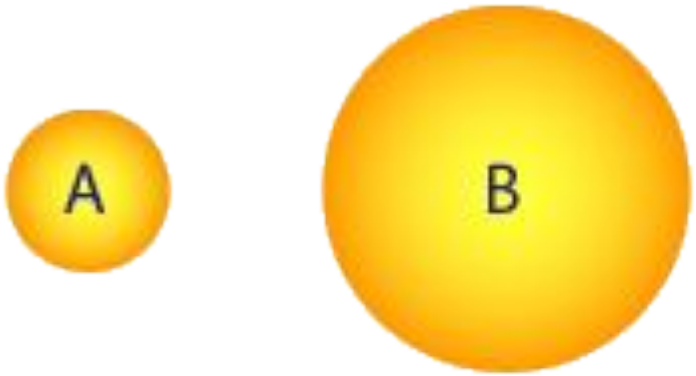

ABSOLUTE MAGNITUDE
#5
#4
#3
ABSOLUTELY Is The Brightest

Luminosity

- Luminosity is the total amount of electromagnetic **energy** (light energy) emitted per unit of **time** by a star
- Luminosity is directly related to **temperature** - the hotter the star the more luminous it is

Luminosity

- Luminosity is also directly related to **power** output - this means that the **size** of the star matters
- Larger stars have a greater **surface area** and therefore a larger luminosity



Stars A and B have the same temperature.
B is bigger than A ∴ has greater surface area.
B ∴ has greater intrinsic luminosity.

Temperature

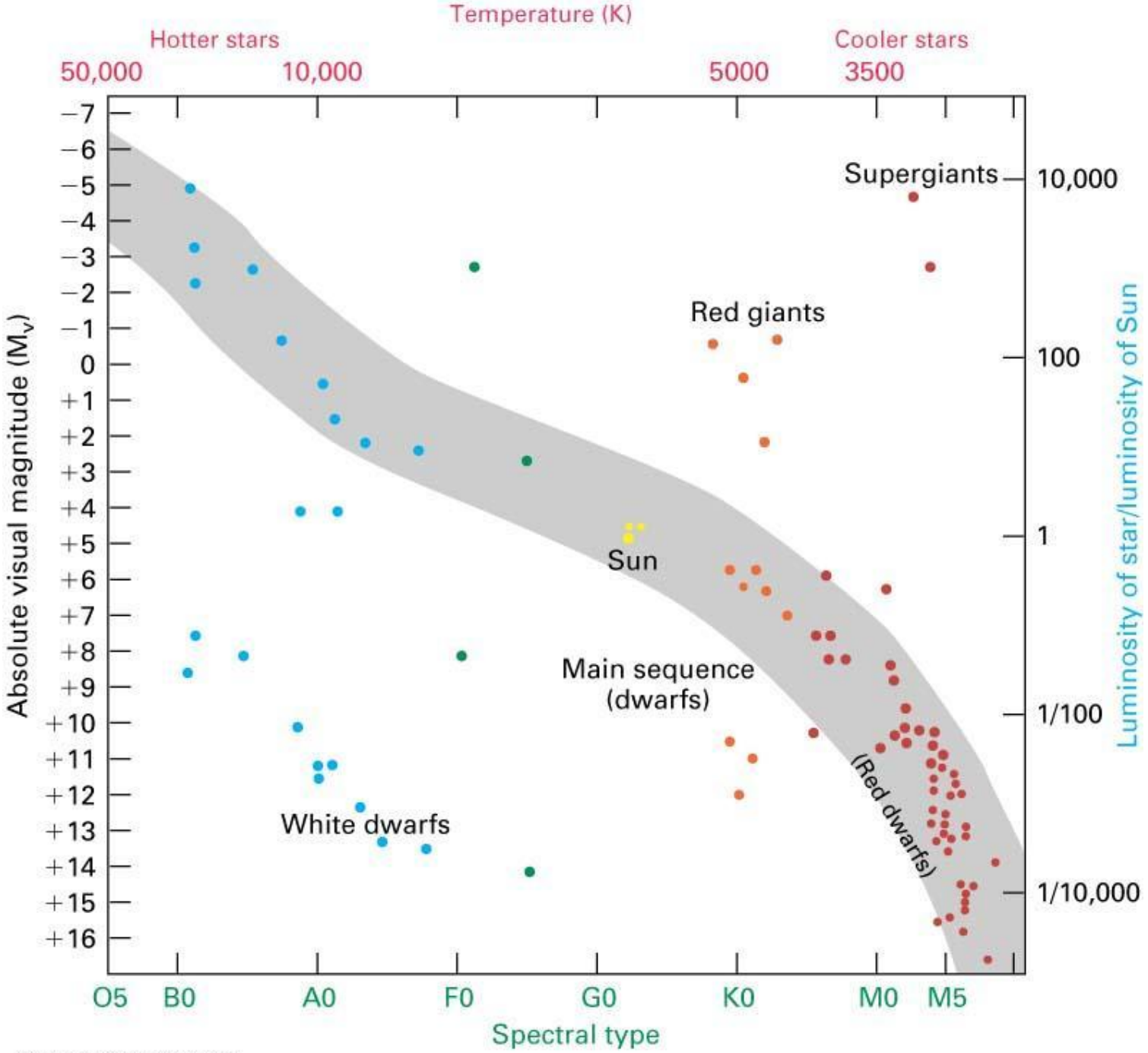
- The **color** of a star is linked to its surface temperature
- The **hotter** the star, the **shorter** the wavelength of light it will emit

Colour	Temperature Range (°C)	Example(s)
Bluish	25 000 – 50 000	Zeta Orionis
Bluish-white	11 000 – 25 000	Rigel, Spica
Whitish	7500 – 11 000	Vega, Sirius
Yellowish-white	6000 - 7500	Polaris, Procyon
Yellowish	5000 – 6000	Sun, Alpha Centauri
Orangish	3500 – 5000	Arcturus, Aldebaran
Reddish	2000 - 3500	Betelgeuse, Antares

HR Diagrams

- A Hertzsprung-Russell (H-R) diagram is a plot of a star's **absolute magnitude** against its **temperature**
- When this is plotted for many stars, it can be seen that most stars lie in a band across the middle of the graph, known as the **main sequence** stars
- About **90%** of all stars are main sequence stars

H-R Diagram



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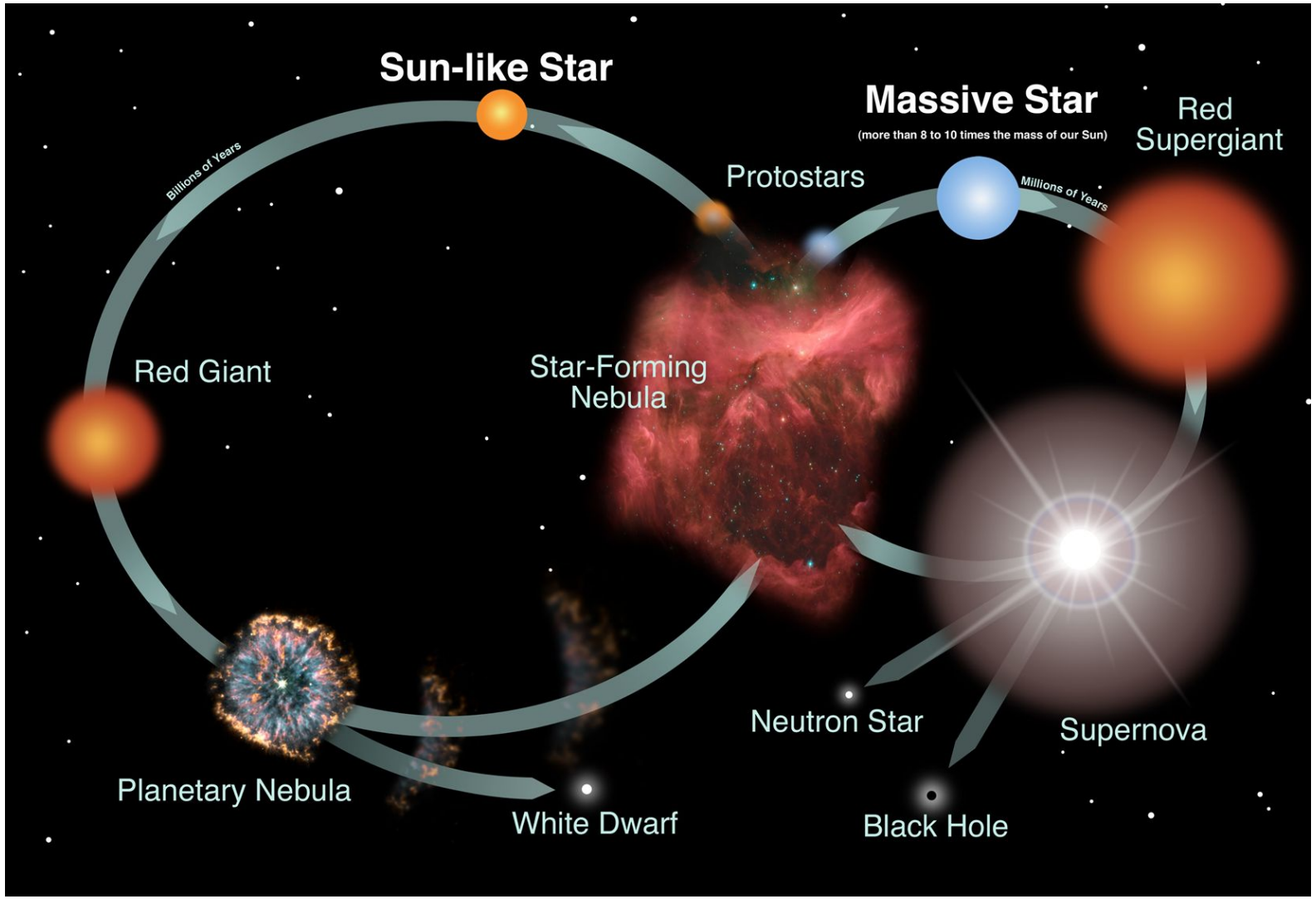
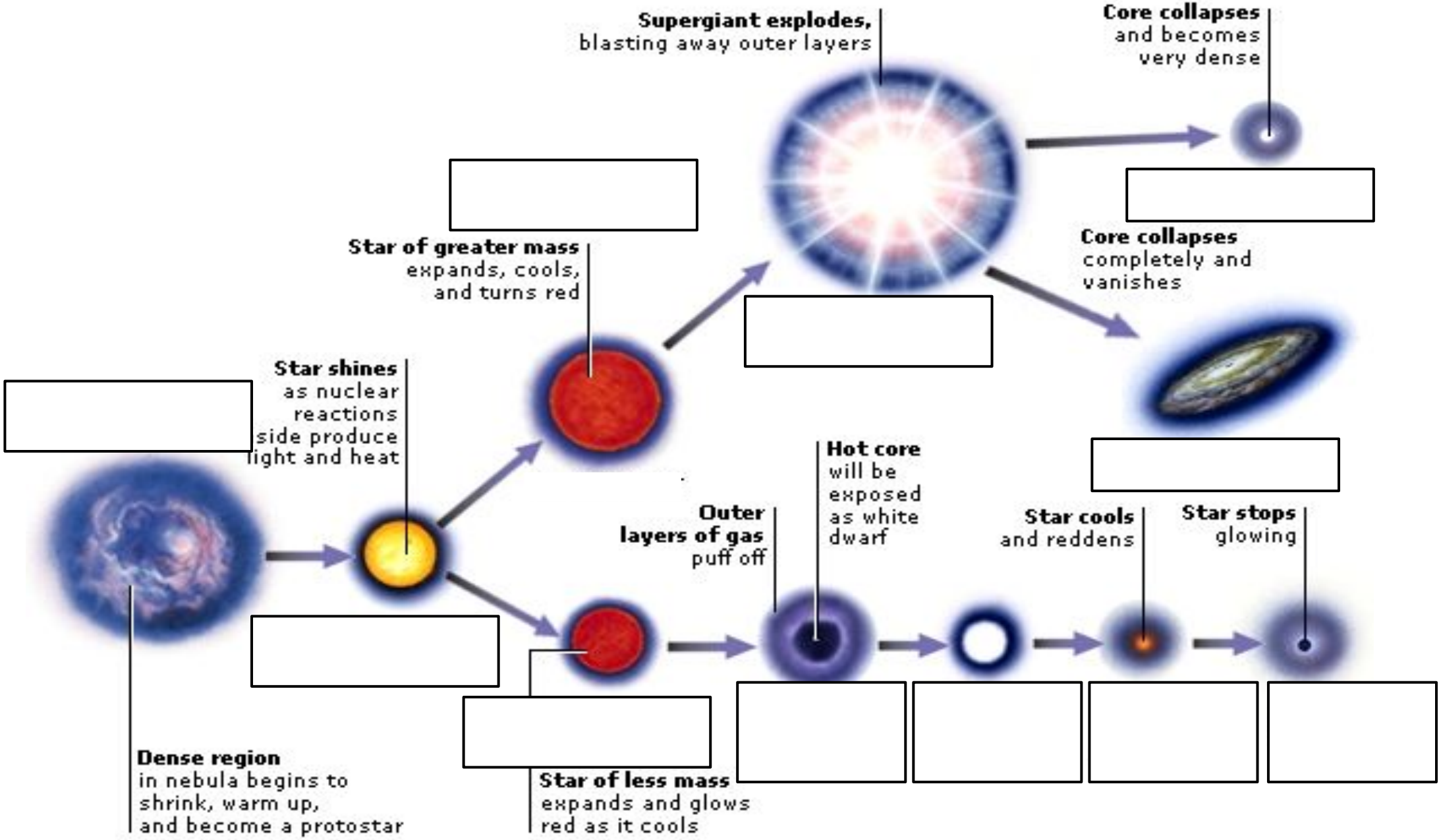
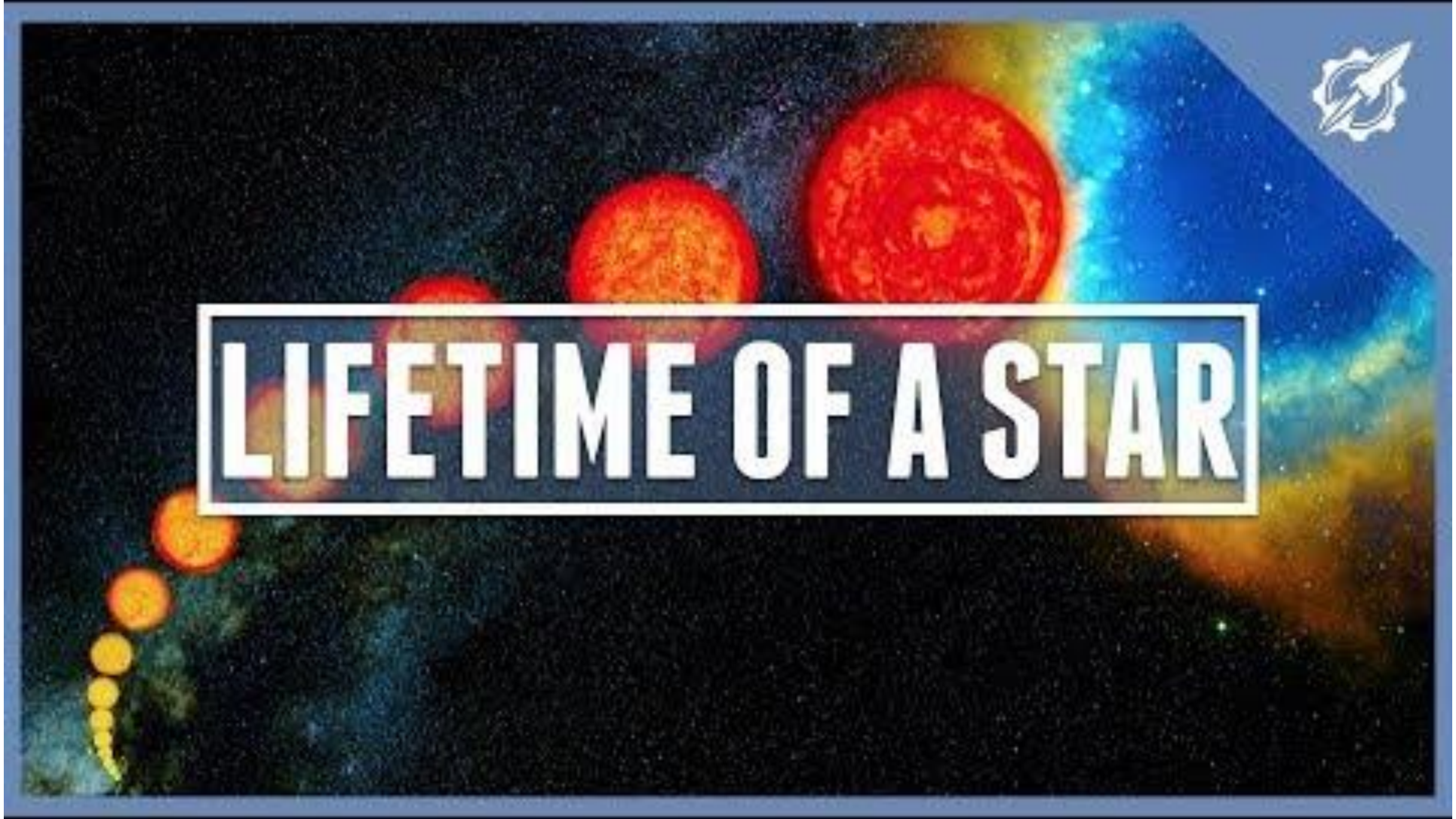
Life Cycle of Stars

"We are all made of stardust" – Carl Sagan



Starts out as a....

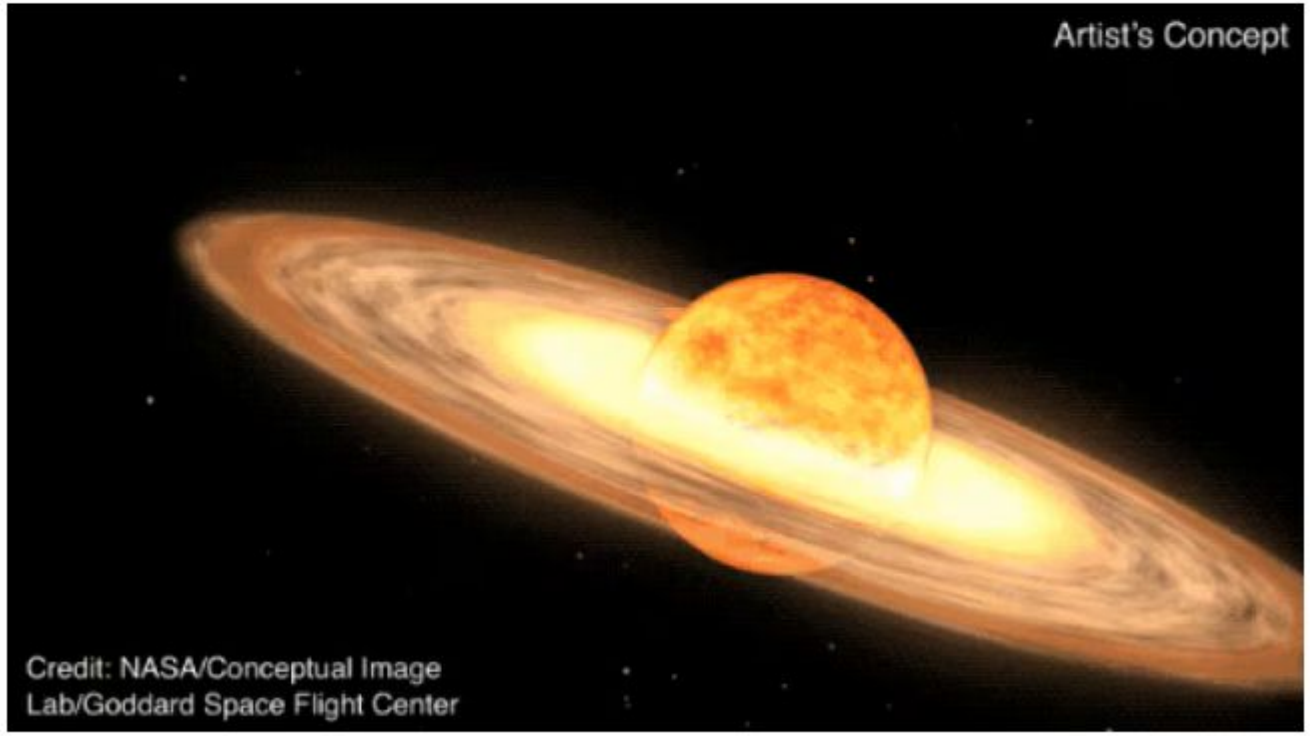



LIFETIME OF A STAR

A 'new' star will light up the sky soon and you can see it for yourself

Recurrent nova last brightened in 1946 and has a recurring cycle of roughly 80 years

Nicole Mortillaro · CBC News · Posted: Apr 17, 2024 4:00 AM EDT | Last Updated: April 17



A binary star system in the constellaion Corona Borealis consists of a white dwarf that pulls material off a neighbouring red giant, leading to exceptionally bright explosions every 79 to 80 years. (NASA's Goddard Space Flight Center)

https://www.cbc.ca/news/science/new-star-t-coronae-borealis-1.7175301